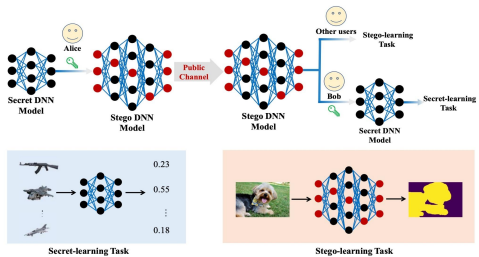


Deep Learning Steganography: Hiding Trained Models Inside Trained Models

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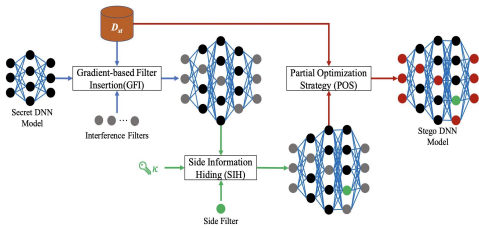
Deep Learning Steganography



DNNs possess representational capacity beyond what is required for a single task. We leverage this redundancy to embed a trained “secret” model inside a publicly deployable “stego” model while preserving the utility of the visible task. The hidden model can later be selectively recovered using a recovery key or filter mechanism.

We also evaluate trade-offs across **capacity**, **cover utility**, and **recovery fidelity**.

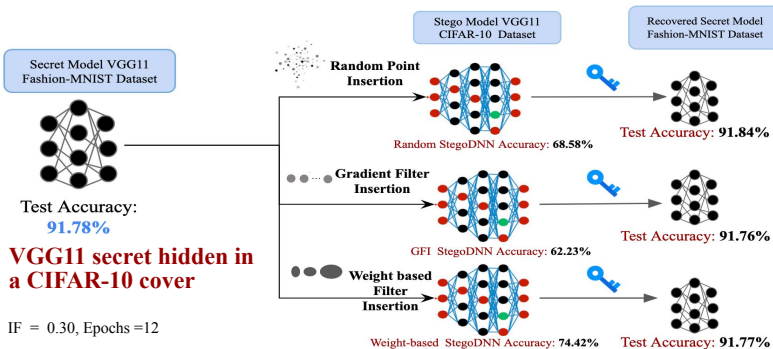
Methodology



Reference

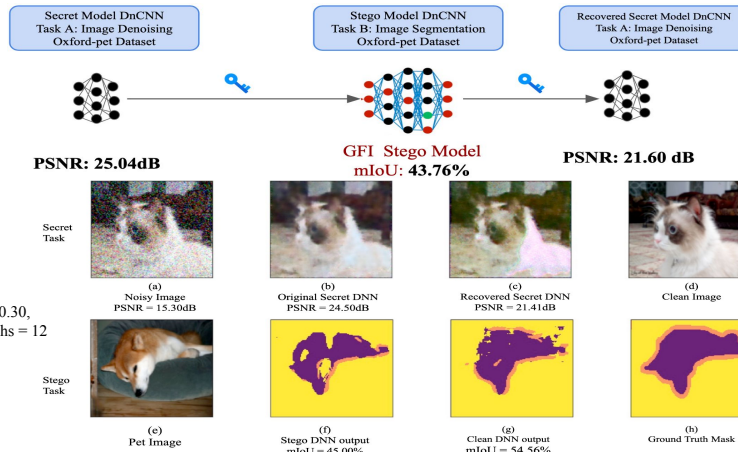
[1] Li, Guobiao, et al. "Towards deep network steganography: from networks to networks." *arXiv preprint arXiv:2307.03444* (2023).

Intra Task Steganography : Same Task (Image Classification)



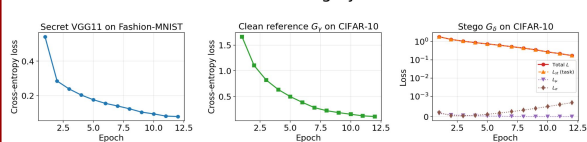
Inter-Task Steganography: 2 Different Tasks

DnCNN denoiser hidden in a segmentation cover

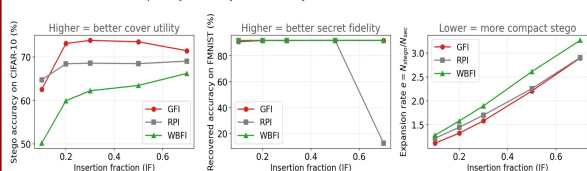


Performance Evaluation

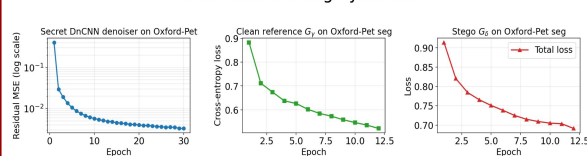
Intra-task Training dynamics



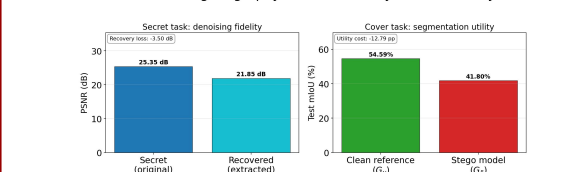
Capacity / Utility / Recovery vs Insertion Fraction (IF)



Inter-task Training dynamics



Inter-task steganography: secret recovery vs cover utility



Gradient-Filter insertion ranks the best. **Intra-task is essentially lossless**. Recovered models closely match the secret model accuracy with drift 7×10^{-6} . **Inter-task pays cost**. Recovery ~ 3.5 dB PSNR; cover utility ~ 12.8 pp mIoU.